Julia Machnio

Øster Voldgade 3
1307 København K
Denmark
+48 533 531 829
juma@di.ku.dk

juliamachnio juliamachnio juliamachnio

Education

Jan. 2024 – Dec. 2026 **Doctor of Philosophy in Computer Science**, *Pioneer Centre for AI, University of Copenhagen*, Copenhagen, Denmark
- PhD studies supervised by Prof. Mads Nielsen and Prof. Mostafa Mehdipour Ghazi

Oct. 2025 – March 2026 **Visiting Researcher**, *ETH Zurich*, Zurich, Swiss
- Visiting Researcher at Biomedical Imaging Computing Group (CVL lab) supervised by Prof. Ender Konukoglu

2022 – 2023 **Master's Degree**, *AGH University of Science and Technology*, Kraków, Poland
- Biomedical Engineering program with specialization in medical informatics and electronics
- Thesis title: MRI data segmentation with deep learning using large-scale computing
- Thesis grade: 5.0/5.0 with honors, Graduation grade: 5.0/5.0
- Supervisors: Prof. Maciej Malawski and Tomasz Pięciak, PhD

2021 – 2021 **Erasmus+**, *GrenobleINP, department Phelma*, Grenoble, France
- Biomedical Engineering Master's program

2018 – 2022 **Bachelor's Degree**, *AGH University of Science and Technology*, Kraków, Poland
- Biomedical Engineering program
- Thesis title: Microstructural brain measures from diffusion MRI using deep learning
- Thesis grade: 5.0/5.0, Graduation grade: 4.65/5.0
- Supervisor: Tomasz Pięciak, PhD

Scholarships and Awards

2024 **1st place**, *XXVth edition of the "AGH Diamonds" contest for best Master's thesis in the category of theoretical theses organized by AGH University of Science and Technology in Kraków.*

2024 **1st place**, *TRUMPF Huettinger's National Competition for the Best Master Thesis defended in 2023*

2024 **2nd place**, *Polish Society for Biomedical Engineering Competition for the Best Master Thesis defended in 2023*

2023 **AGH University Scholarship**, *Top-Performed Student, Class of 2022*

2023 **3rd place**, *TRUMPF Huettinger's National Competition for the best Bachelor Thesis defended in 2022*

2022 **1st place**, *IEEE and AGH faculty of Electronics, Robotics, Informatics, and Biomedical Engineering's Competition for the best Bachelor Thesis*

2022 **1st place**, *IEEE and 4 Science Institute's National Contest for the Most Scientific Bachelor Thesis*

2022 **3rd place**, *IEEE, 4 Science Institute and GovTech Poland's National Contest for best AI related Bachelor Theses*

2022 **SANO Master's scholarship**, *SANO Centre for Computational Medicine's scholarship for Master Thesis*

2019, 2020, 2021, 2022 **AGH Rector's scholarship**
Scholarship awarded each year to the top 10% of students with the highest grades.

Publications

ECCV 2026 **A Mechanism-Driven Theory of Phase Transitions in Active Learning**
J Machnio, M Nielsen, MM Ghazi
European Conference on Computer Vision (ECCV), 2026

EMA4MICCAI 2026 **Oral** **How Many Labels Are Enough? ALDA: Active Learning Deployment Advisor for Medical Image Classification**
J Machnio, M Nielsen, MM Ghazi
2nd MICCAI Workshops on Efficient Medical AI, 2026

ArXiv 2026 **Towards Brain MRI Foundation Models for the Clinic: Findings from the FOMO25 Challenge**
A Munk, ..., **J Machnio** et al.
ArXiv, 2026

SPIE 2026 **Deep Learning-Based Regional White Matter Hyperintensity Mapping as a Robust Biomarker for Alzheimer's Disease**
J Machnio, M Nielsen, MM Ghazi
SPIE Medical Imaging Conference, 2026

SPIE 2026 **MRI Embeddings Complement Clinical Predictors for Cognitive Decline Modeling in Alzheimer's Disease Cohorts**
N Putera, D Vilet Rodrıguez, N Videcrantz, **J Machnio**, MM Ghazi
SPIE Medical Imaging Conference, 2026

ICCV 2025 **To Label or Not to Label: PALM – A Predictive Model for Evaluating Sample Efficiency in Active Learning Models**
J Machnio, M Nielsen, MM Ghazi
International Conference on Computer Vision (ICCV), 2025

SHW 2025 **Oral** **Towards Scalable and Robust White Matter Lesion Localization via Multimodal Deep Learning**
J Machnio, M Nielsen, MM Ghazi
2nd Sorbonne-Heidelberg Workshop on AI in medicine: Machine Learning for multi-modal data, 2025

ArXiv 2025 **A large-scale heterogeneous 3D magnetic resonance brain imaging dataset for self-supervised learning**
A Munk, S Cerri, J Ambsdorf, **J Machnio** et al.
ArXiv, 2025

NLDL 2025 **Oral** **Deep Learning for Localization of White Matter Lesions in Neurological Diseases**
J Machnio, M Nielsen, MM Ghazi
Northern Lights Deep Learning Conference (NLDL), 2025

ArXiv 2024 **Yucca: A deep learning framework for medical image analysis**
SN Llambias, **J Machnio**, A Munk, J Ambsdorf, M Nielsen, MM Ghazi
ArXiv, 2025

MELBA 2024 **MICCAI-CDMRI 2023 QuantConn Challenge Findings on Achieving Robust Quantitative Connectivity through Harmonized Preprocessing of Diffusion MRI**
NR Newlin et al.
Machine Learning for Biomedical Engineering Journal (MELBA), 2024

MICCAI 2024 **Introducing QuantConn: Overcoming Challenging Diffusion Acquisitions with Harmonization**
NR Newlin et al.
International Workshop on Computational Diffusion MRI (MICCAI-CDMRI), 2024

Teaching

2024 – 2025 **Deep Learning Course**, *University of Copenhagen*, Teaching Assistant
Teaching assistant for helping with student assignments and grading

Reviewing

Journal of Alzheimer's Disease, Journal of Biochemistry and Bioinformatics, MICCAI26, ECCV26, Neurips26

Academic Activities

2026 **Oral Presentation**, *Learning Theory Workshops*, Copenhagen, Denmark

2026 **FOMO26 Challenge Co-Organizer**, *29th International Conference on Medical Image Computing and Computer Assisted Intervention*, Strasburg, France

2025 **FOMO25 Challenge Co-Organizer**, *28th International Conference on Medical Image Computing and Computer Assisted Intervention*, Daejon, South Korea

2024 **Poster Presentation**, *Danish Digitalization, Data Science and AI Conference*, Nyborg, Denmark

2024 **Poster Presentation**, *Summer School on Biomedical Image Analysis - from acquisition to fairness and bias*, Svendborg, Denmark

2024 **Poster Presentation**, *Northern Lights Deep Learning Winter School*, Trømso, Norway

2023 **Poster Presentation**, *International Computer Vision Summer School*, Sicily, Italy

2023 **Poster Presentation**, *International Neuro AI Summer School*, Lipari, Italy

2023 **Abstract Presentation**, *XVth HPC Users' Conference - KUKDM*, Zakopane, Poland

2022 **Poster Presentation**, *XXXIst National Conference "Human and his environment"*, Kielce, Poland

Work Experience

Sep.2023 – Dec.2023	**Junior Scientific Researcher**, *Sano Centre for Computational Medicine*, Kraków, Poland
Jul.2022 – Dec.2023	**ServiceNow Developer**, *Capgemini Sp.z.o.o*, Kraków, Poland
Aug.2021– Jul.2022	**Junior ServiceNow Developer**, *Capgemini Sp.z.o.o*, Kraków, Poland

Personal and Technical Skills

Languages	Polish (Native), English (Fluent), French (Intermediate), German (Intermediate)
Programming	Python, MATLAB, LaTeX, C/C++, Labview
Deep Learning	PyTorch, TensorFlow
Medical Related	FSL, FreeSurfer
Other	Bash, SLURM, Git, Linux, HPC Clusters